Q02: Is Next-Token Generation Intelligence?

Turn a label into testable capability claims

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The student question

Q02
Is next-token generation intelligence? How did intelligence emerge from deep learning?

- Separate a philosophical label from observable behavior.
- Ask what an experiment would have to show.

One word, many possible claims

- Task performance
- Flexible generalization
- Planning and learning
- Social understanding or consciousness

Do not let a single word hide which claim is being made.

The objective does not settle the label

$$\min_{\theta} -\sum_{t} \log p_\theta(x_{t+1} \mid x_{\leq t})$$

- A next-token objective can yield useful task behavior.
- It neither proves nor disproves every sense of "intelligence."
- Architecture, data, post-training, prompting, and tools matter too.

Make a claim testable

Too broad: "The model reasons."

Testable: On held-out, adversarially paraphrased problems, it achieves a stated score and its verifier-checked artifacts satisfy a stated property.

Specify: task distribution, metric, baselines, and failure analysis.

What scale results can show

- Performance can improve with data, compute, and capacity.
- Some benchmark curves look threshold-like.
- A threshold may be algorithmic, metric-induced, or both.

Wei et al. (2022); Schaeffer et al. (2023)

Three bad inferences

1. Fluency $\Rightarrow$ understanding
2. One benchmark $\Rightarrow$ general intelligence
3. One failure $\Rightarrow$ no capability

Each shortcut discards the task, distribution, or alternative explanation.

Mini experiment

- Show a simple rule through examples.
- Hold out combinations that distinguish a rule from memorization.
- Compare a rule-based solver with a lookup baseline.
- Report exact successes and failures.

The exercise tests a narrow capability, not intelligence as a whole.

Takeaway

Scientific posture
Treat "intelligence" as a family of claims. State the behavior precisely, test it under meaningful distribution shift, and keep conceptual questions separate from measurement claims.

References: Turing (1950); Brown et al. (2020); Wei et al. (2022); Schaeffer et al. (2023); Bender et al. (2021).